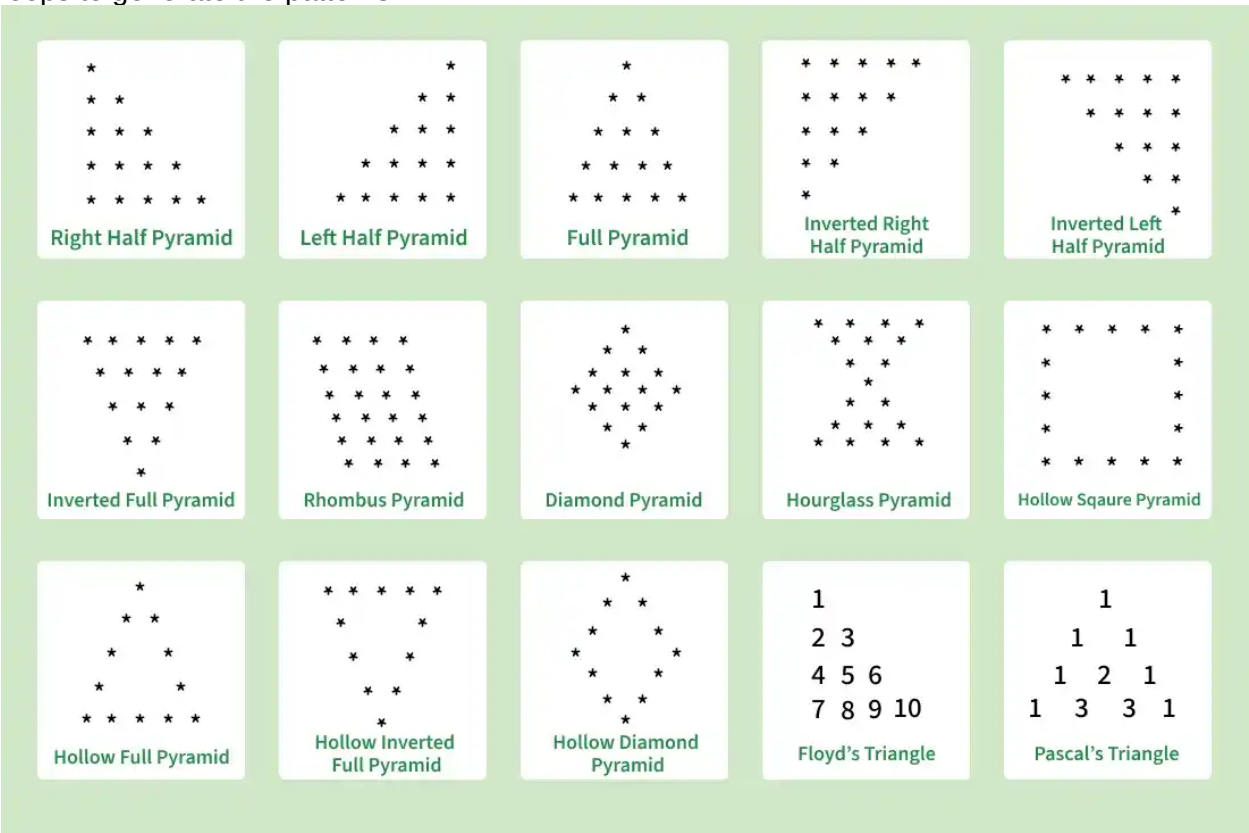
ASSIGNMENT No. 04

Looping Control Structures: *Implement programs using for, while, and do-while loops to generate number patterns, check prime numbers, compute factorial, Fibonacci series, Armstrong numbers, and multiplication tables.*

1. WAP using loop (for, while, do-while):
   • Factorial of number
   • Whether no. is prime or not
   • X raised to power n
   • Armstrong numbers between 1 to 500
   • Range of set of entered nos. by finding smallest and largest
   • Multiplication table
   • HCF of two numbers
   • GCD of two numbers\
   • Generate Fibonacci series
2. Write a program to read in numbers until the number -999 is encountered. The sum of all numbers read until this point should be printed out.
3. Keep reading in integers until one integer larger than 100 is not input. For example: 4, 90, 35, 78, 110 QUIT! (since 110 is bigger than 100).
4. Read in 20 integers and count the even numbers.
5. Write programs that print the following patterns separately, one after the another. Use for loops to generate the patterns.



6. A sensor records the room temperature every 4 hours for 24 hours. Write a C program using for loop to input all the temperature readings and then display the highest temperature recorded in the day.
7. Using the while loop, write a C program to print the Fibonacci sequence up to N numbers, where N is entered by the user at run time. (NOTE: The Fibonacci sequence is a series of numbers in which each number is the sum of the two preceding ones. The starting two preceding numbers are 0 and 1. The Fibonacci sequence till 10 numbers are 0, 1, 1, 2, 3, 5, 8, 13, 21, 34)
8. You are driving a car with N litres of fuel. With 1 litre of fuel, you can cover 5 kms distance. Write a C program using a while loop to calculate how many kilometres you can travel until the fuel becomes 0, and print the fuel left after every 10 kilometres. (For example: If N = 20 Liters, and you travelled 10 kms, fuel left 18 Liters, then travelled 10 kms, fuel left 16 Liters and so on)
9. At a cinema hall, N tickets are available. Each person in the queue buys between 1 to 4 tickets. Write a C program using a while loop to simulate the sale of tickets until all tickets are sold and display the number of tickets left after each purchase. The starting N tickets should be entered by the user at run time (For example: If N = 10, and you entered tickets required = 4 then tickets available = 6; you entered tickets required = 3 then tickets available = 3, and soon).
10. You deposit ₹10,000 in a bank account with 5% annual interest. Each year, the bank adds interest to your balance. Write a C program using a while loop to calculate how many years it will take for your balance to exceed ₹15,000. (For example: Year 1, ₹10,500.00; Year 2, ₹11,025.00; Year 3, ₹11,576.25, and so on).
11. Write a C program using a do-while loop for a number guessing game. The computer has a secret number (for example, 49). Ask the user to guess the number until they get it right. After each wrong guess, tell them if their guess was too high or too low.
12. Write a C program using a do-while loop that asks the user to enter their weight (in kg) and height (in meters). Make sure both values are positive. If not, keep asking until valid numbers are entered. Then, calculate the Body Mass Index (BMI) using the formula: BMI = weight (Kg)/height(m2). After that, tell the user if they are overweight (BMI > 25) or not.